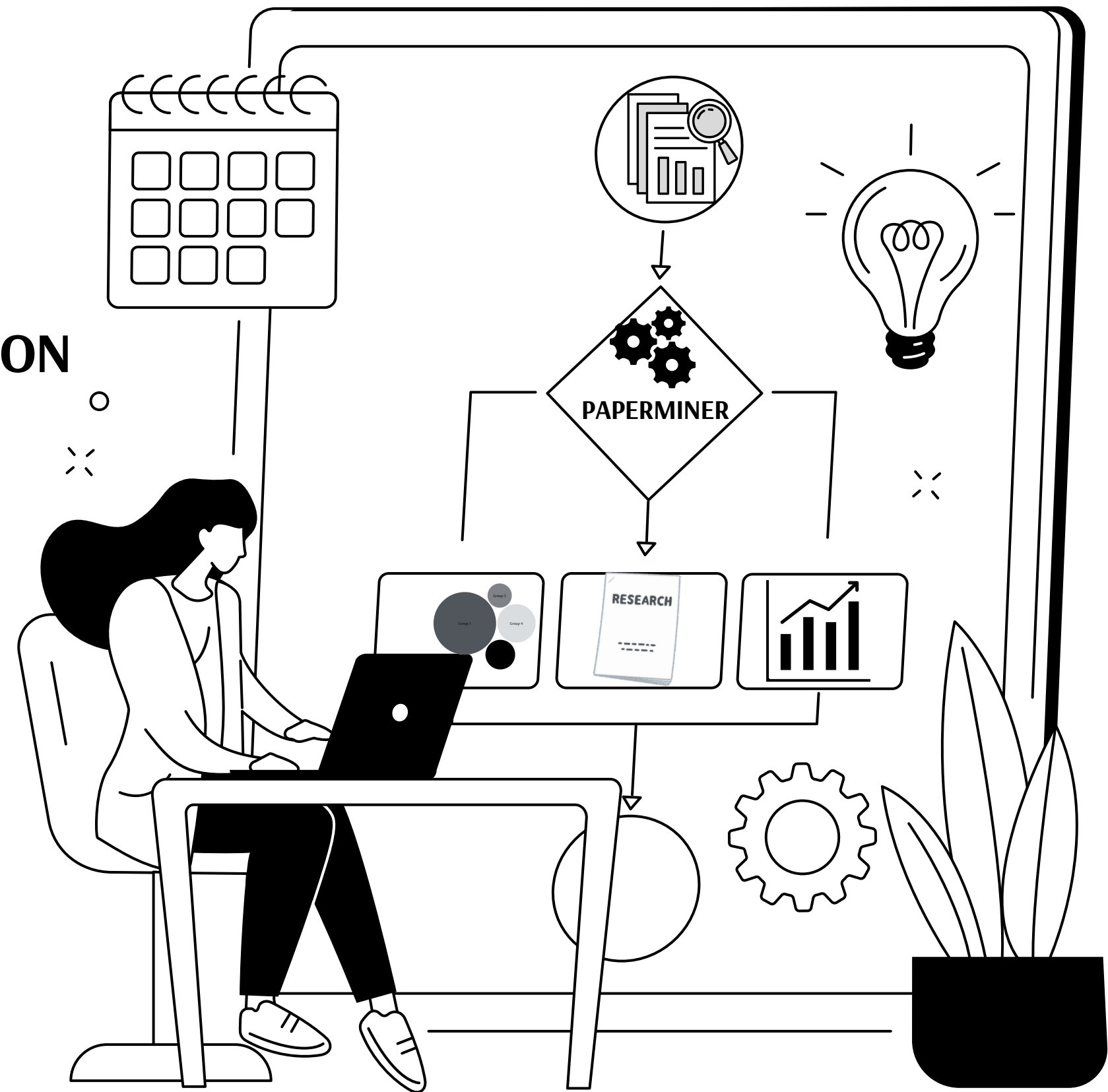


PAPERMINER

SCIENTIFIC LITERATURE INTELLIGENCE SOLUTION

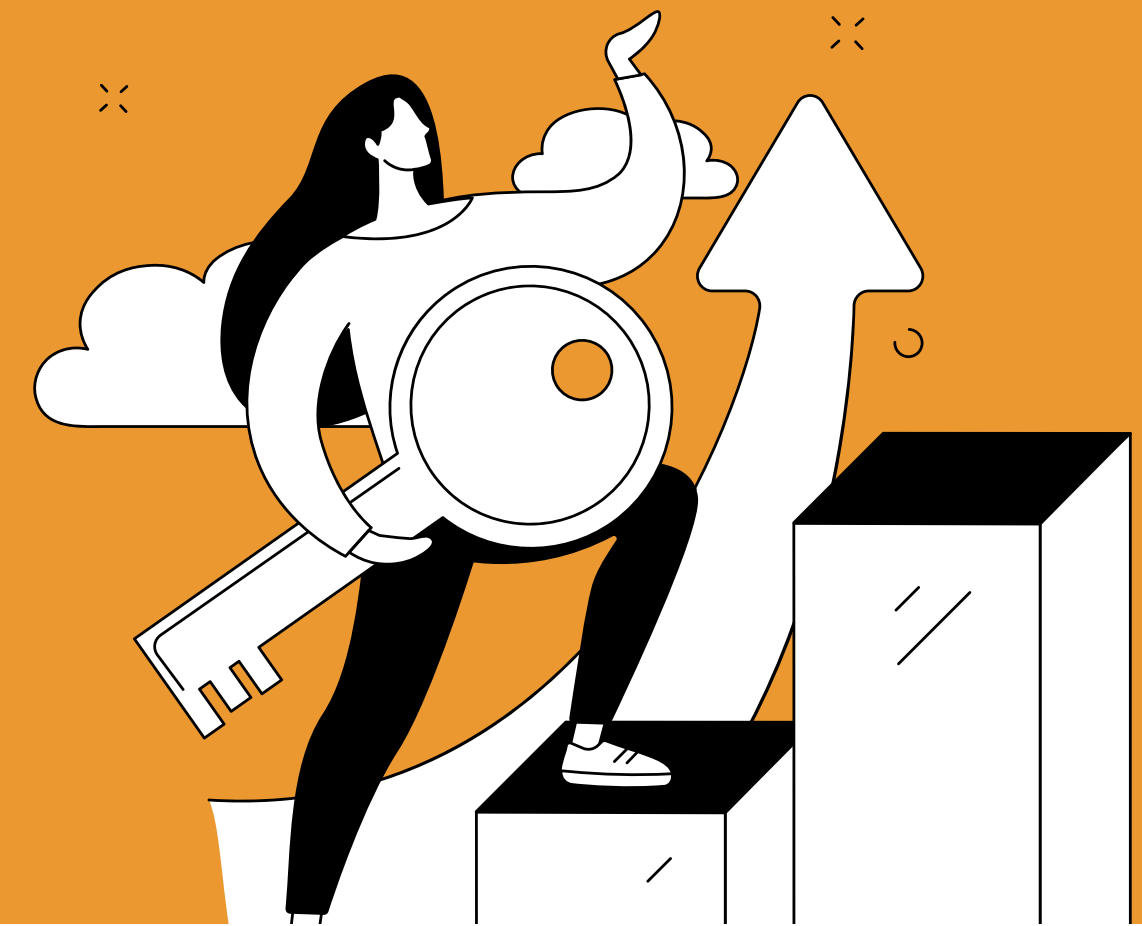


Built and Presented by
Sanghamitra Matta

SOLUTION TO WHAT?

Scientific publication output doubles roughly every 9–15 years, with millions of papers published annually.

Researchers often struggle to manually track, read, and synthesize all relevant literature in their field



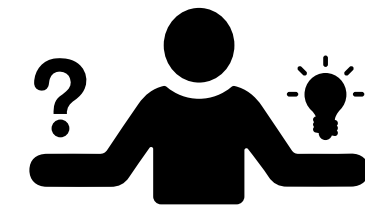
To be up-to-date and Accelerate Discovery

- Can scan and filter thousands of journals
- Surface the most relevant papers based on semantic similarity
- Can dramatically shortens the "time to insight", accelerating the entire scientific cycle.



Can be Predictive and can give Meta-Science Insights

- Detect "hot" or rapidly evolving research topics
- Identify potential collaborators based on complementary work, even across fields.
- Highlight important questions that are under-studied



Can democratize Access and Bridge the Knowledge Gap

- Lower the entry barrier for new researchers entering a field with curated, synthesized knowledge
- Assist non-native English speakers with translation and summarization.

WHERE IT FITS?

Really anywhere where you need research!

R&D - Academia

- Literature scans and state-of-the-art briefs
- Compare methods and extract key findings
- Draft citations for reports and papers

Policy / Legal / Compliance

- Find relevant passages fast
- Generate summaries with traceable quotes
- Create audit-friendly evidence trails

Education

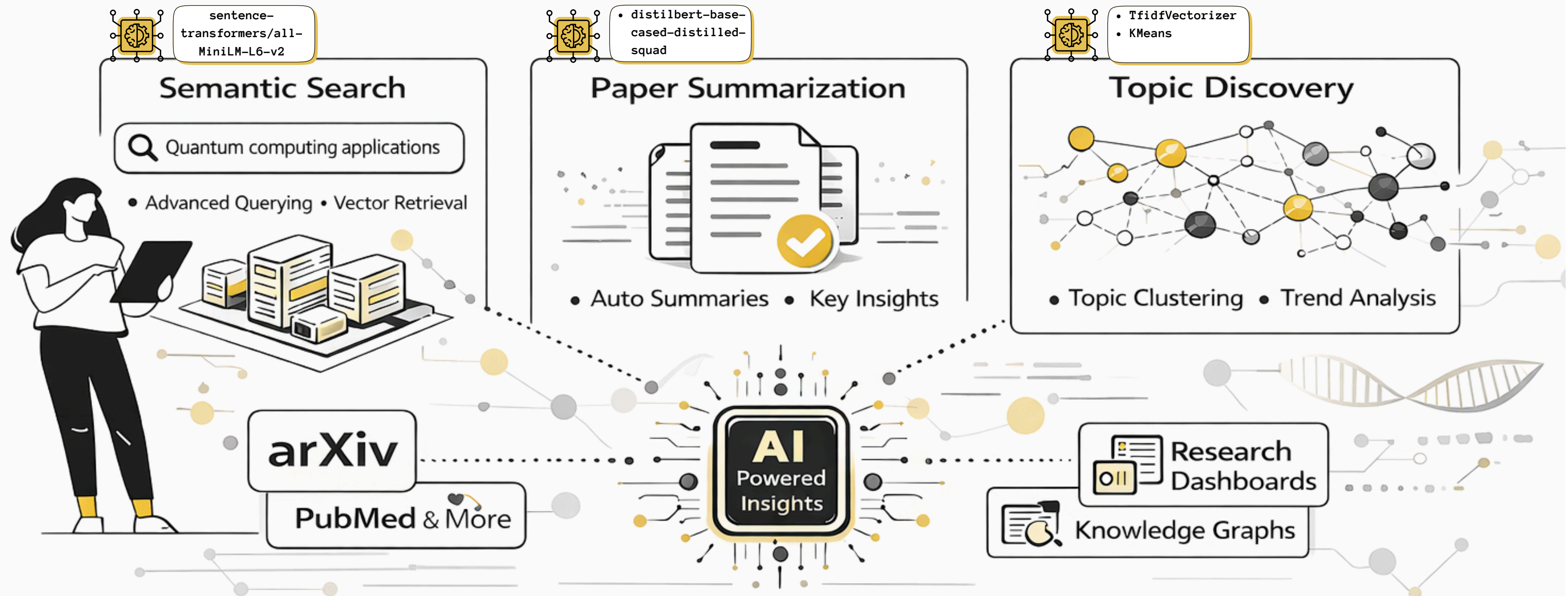
- Turn dense docs into simple explanations
- Onboard new team members faster
- Reusable “one-pagers” based on reliable sources



Paperminer

Literature Intelligence Platform

Semantic search, paper summaries, and topic discovery for **scientific** literature



Stay Current | Save Time | Gain Insights

PRODUCT OVERVIEW

LITERATURE SEARCH - SEMANTIC & KEYWORD

- Search papers using natural language queries
- Results ranked by semantic relevance (embeddings)

TOPIC DISCOVERY + EXPLORATION

- Automatically clusters papers into topics
- Lets users browse “what’s happening in this area”
- Browse the related paper

Q & A

Ask interactive questions over a set of papers or a single paper . Answers are retrieved and compiled from the top-matching papers, with cited sources.



WHATS UNDER THE HOOD ?

- Semantic Search / Retrieval: sentence-transformers/all-MiniLM-L6-v2 (used for tasks like clustering or semantic search)
Fast CPU embeddings for semantic similarity + top-K retrieval(RAG).
- Topic Discovery: TF-IDF (TfidfVectorizer) + KMeans (k=8)
Lightweight, interpretable topics for browsing + filtering.
- Q&A :(RAG Based) distilbert-base-cased-distilled-squad
Answers are extracted from retrieved text to reduce hallucinations.
- Summarization: sshleifer/distilbart-cnn-12-6
Small + runs on CPU
Cached with lru_cache(maxsize=1) so Streamlit won't reload the model repeatedly



GUARDRAILS

(⚙️) This tool summarizes and answers questions based on the retrieved papers only. It shows citations for verification. If information isn't in the retrieved sources, it will say so. It does not provide medical/legal advice, and it will not fabricate citations or results

- Scientific integrity - Q&A must answer only from retrieved paper text/snippets. No fabrication rule
- Scope - what the tool is / isn't
- Source Transparency
- Security
- Copyright & licensing
- UX
- Evaluation - Automated checks, Human review loop, Metrics to track



MEET OUR TEAM



???



Sanghamitra Matta



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